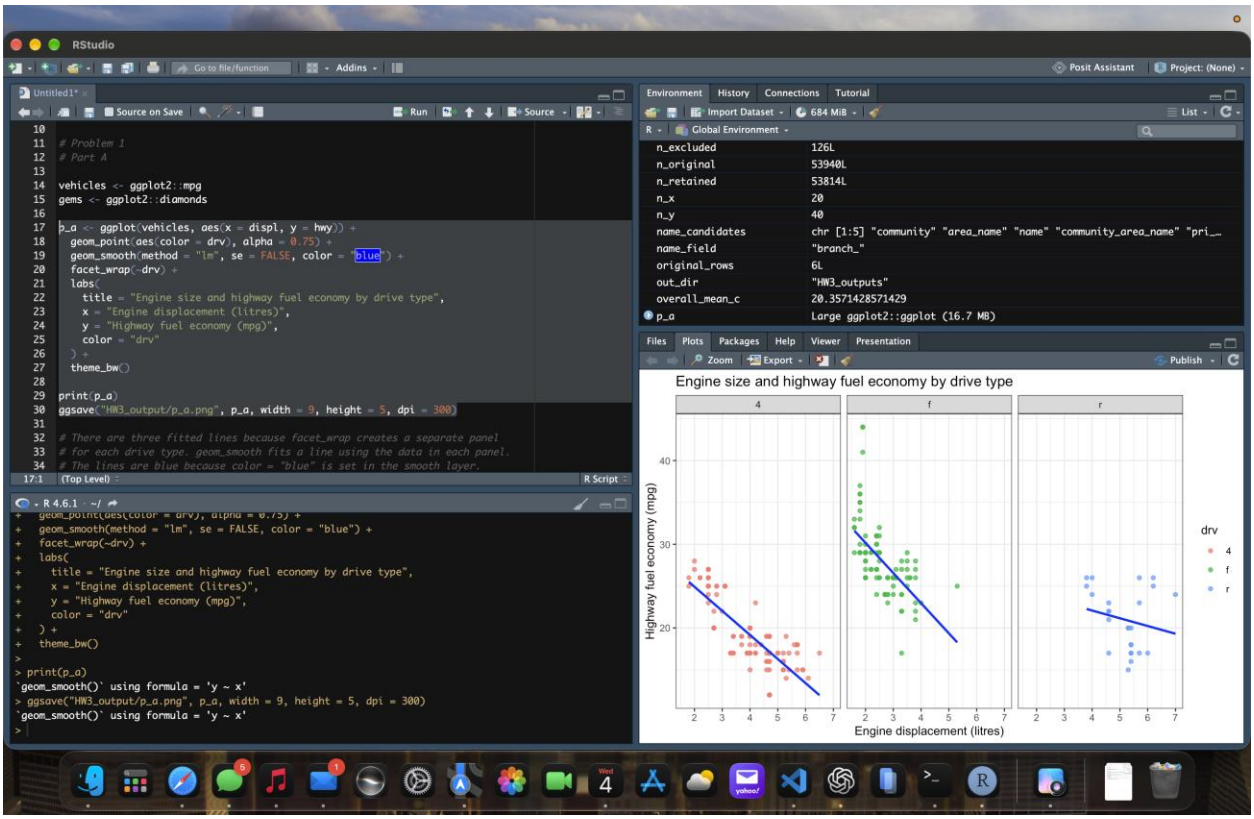
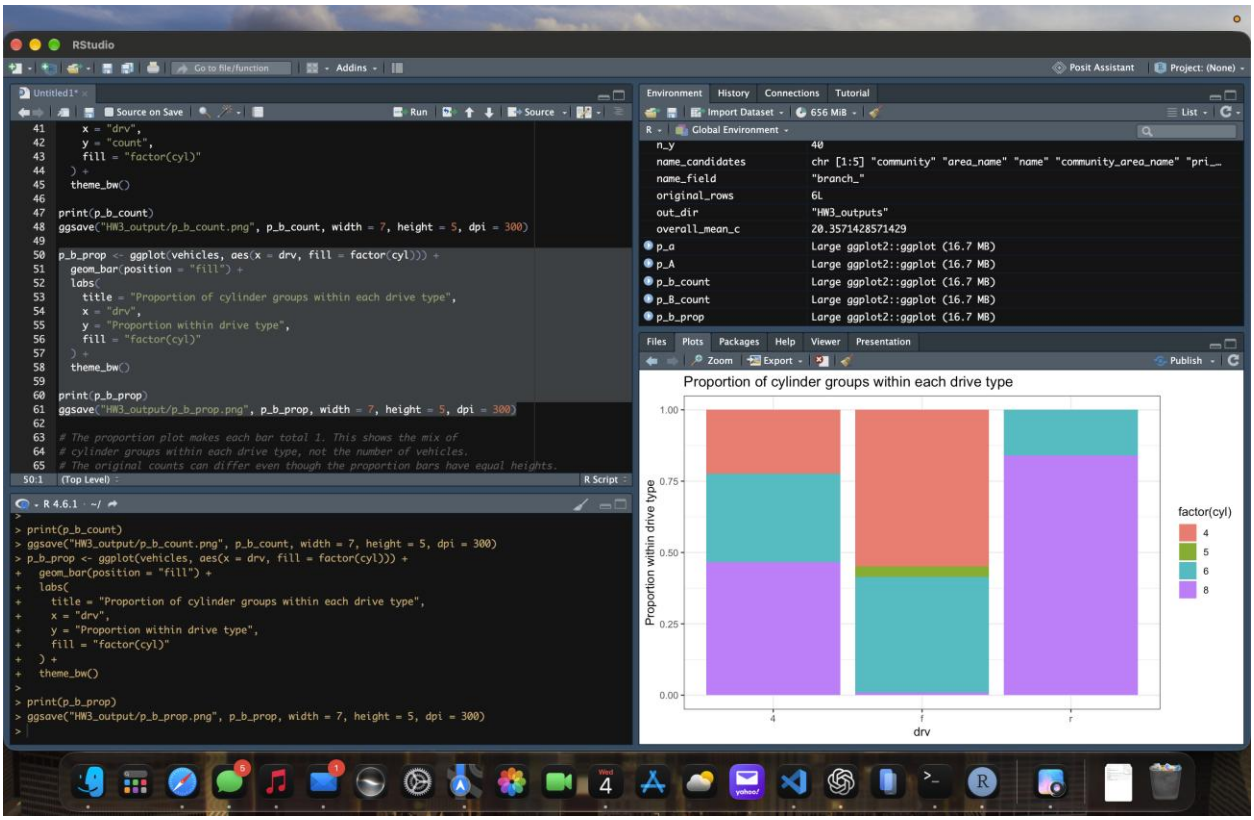
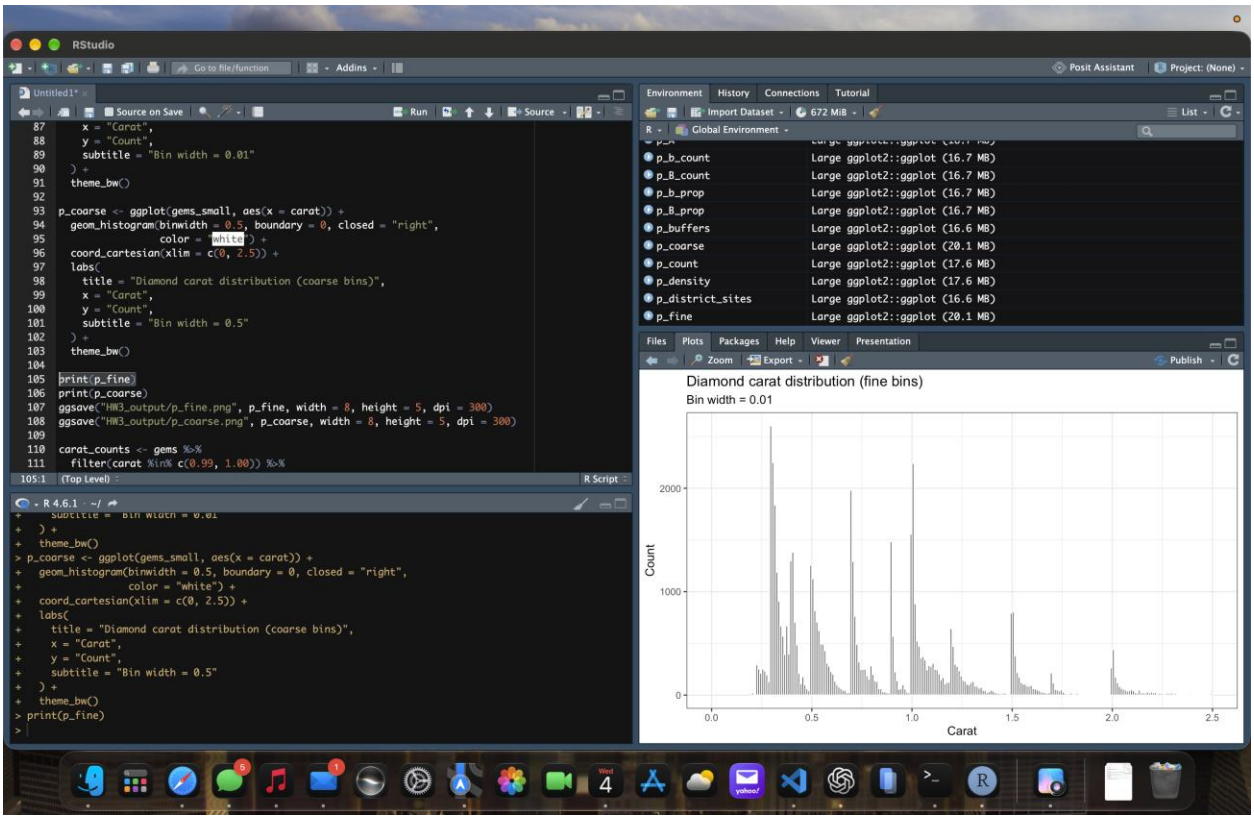
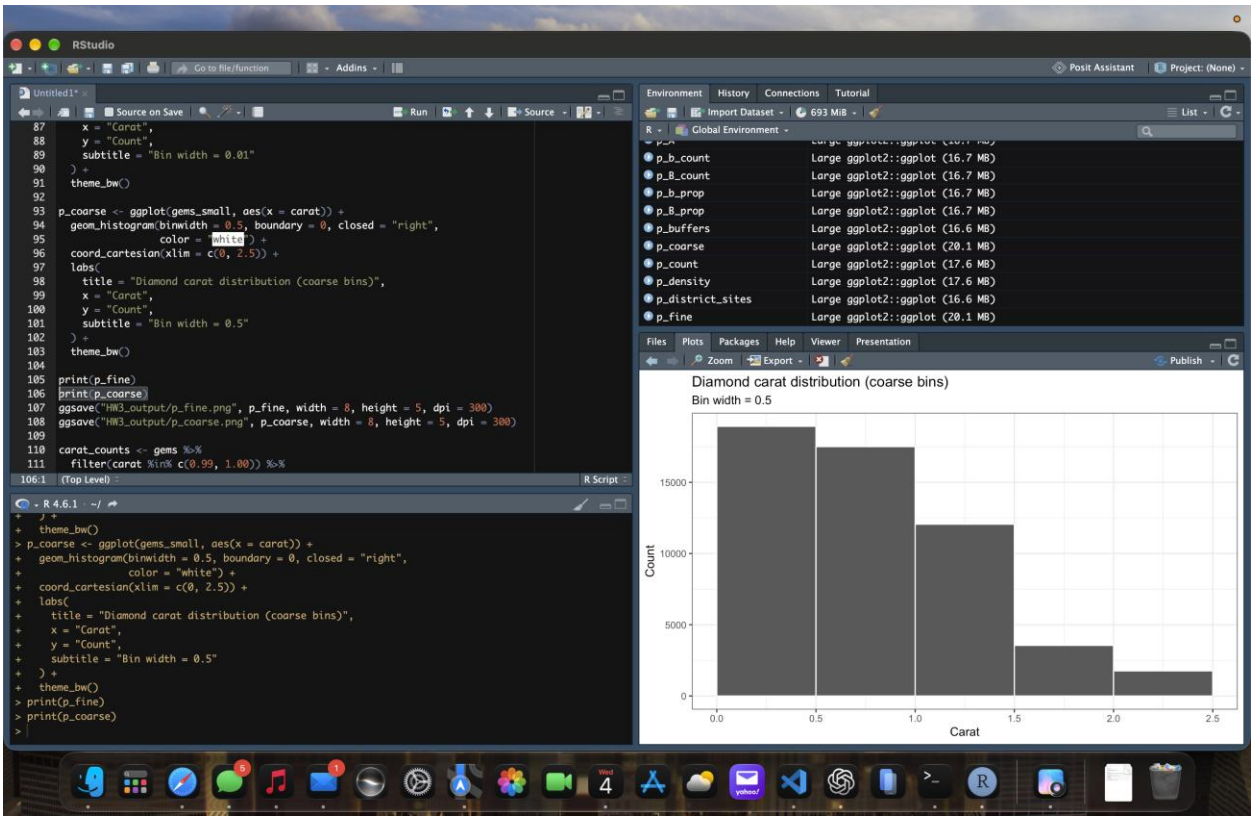
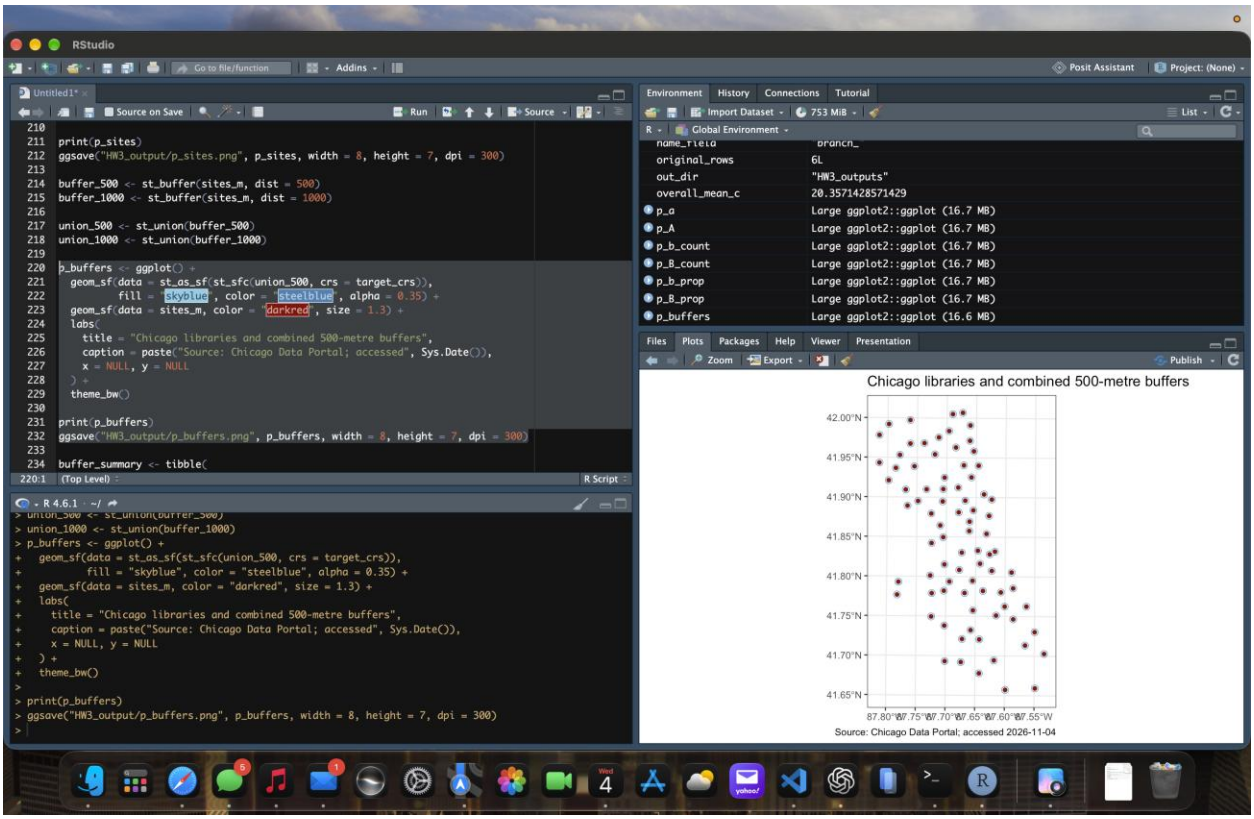
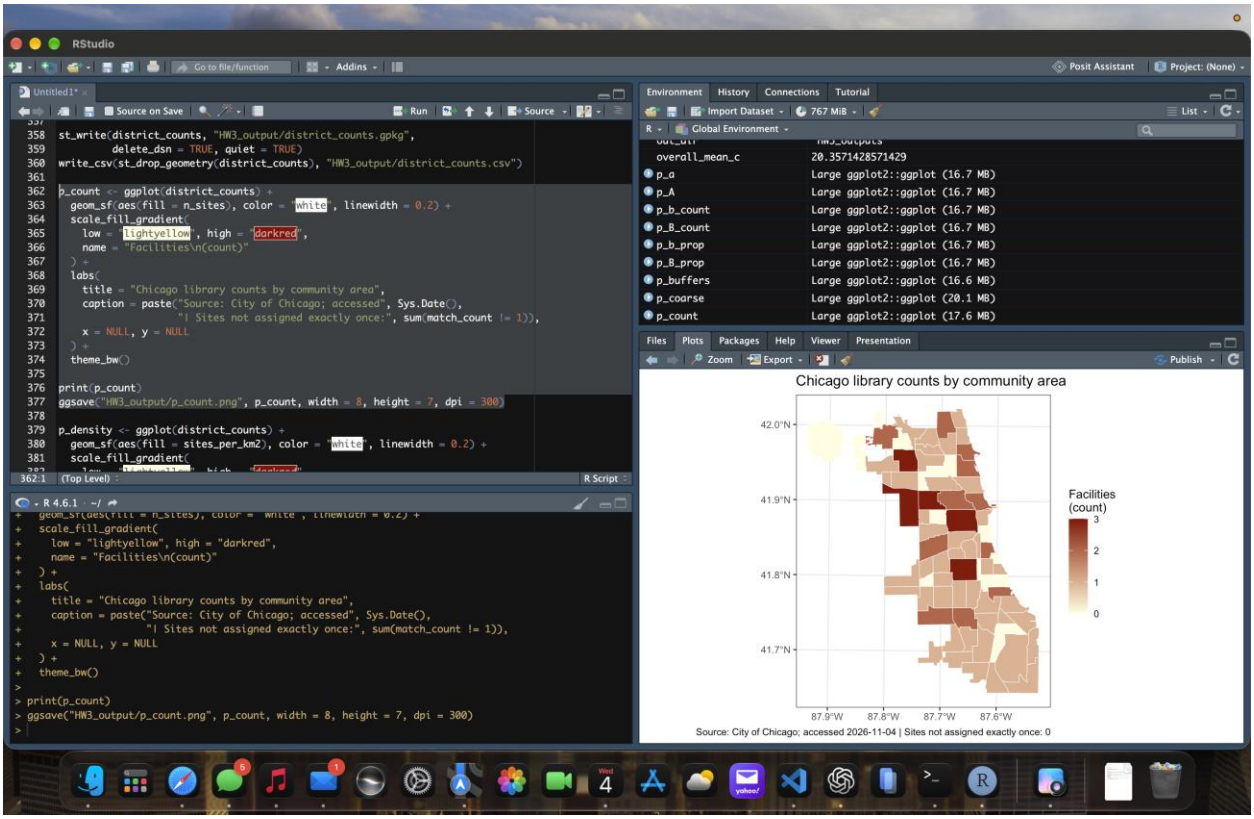

RStudio
Untitled1*
Problem 1
Part A
vehicles <- ggplot2::mpg
gems <- ggplot2::diamonds
p_a <- ggplot(vehicles, aes(x = displ, y = hwy)) +
geom_point(aes(color = drv), alpha = 0.75) +
geom_smooth(method = "lm", se = FALSE, color = "blue") +
facet_wrap(~drv) +
labs(
title = "Engine size and highway fuel economy by drive type",
x = "Engine displacement (litres)",
y = "Highway fuel economy (mpg)",
color = "drv"
) +
theme_bw()
print(p_a)
ggsave("HW3_output/p_a.png", p_a, width = 9, height = 5, dpi = 300)
There are three fitted lines because facet_wrap creates a separate panel
for each drive type. geom_smooth fits a line using the data in each panel.
The lines are blue because color = "blue" is set in the smooth layer.
17:1 (Top Level) R Script
R 4.6.1 · ~/
`geom_smooth()` using formula = 'y ~ x'
> ggsave("HW3_output/p_a.png", p_a, width = 9, height = 5, dpi = 300)
`geom_smooth()` using formula = 'y ~ x'
Environment History Connections Tutorial
Import Dataset 684 MiB
Global Environment
n_excluded 126L
n_original 53940L
n_retained 53814L
n_x 20
n_y 40
name_candidates chr [1:5] "community" "area_name" "name" "community_area_name" "pri…
name_field "branch_"
original_rows 6L
out_dir "HW3_outputs"
overall_mean_c 20.3571428571429
p_a Large ggplot2::ggplot (16.7 MB)
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Engine size and highway fuel economy by drive type
4
f
r
Highway fuel economy (mpg)
Engine displacement (litres)
drv
4
f
r

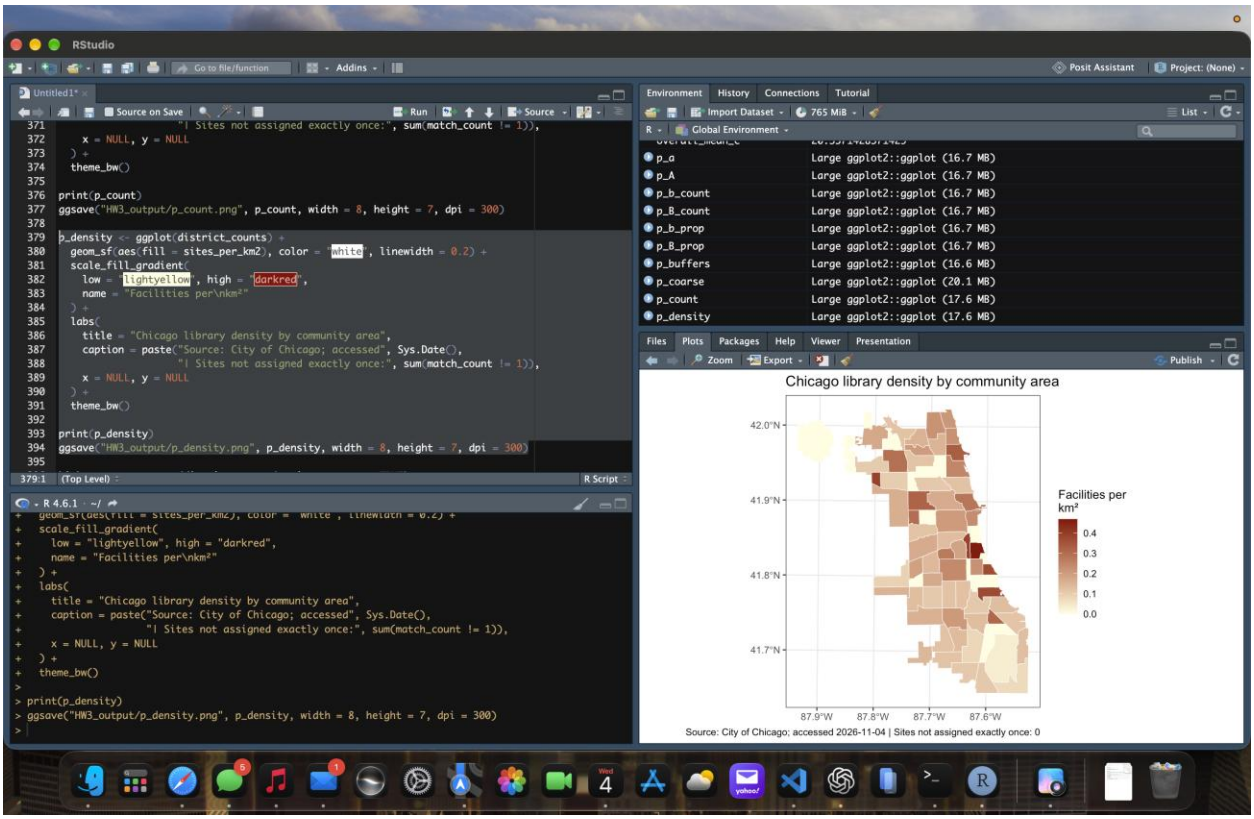

RStudio
Untitled1*
x = "drv",
y = "count",
fill = "factor(cyl)"
) +
theme_bw()
print(p_b_count)
ggsave("HW3_output/p_b_count.png", p_b_count, width = 7, height = 5, dpi = 300)
p_b_prop <- ggplot(vehicles, aes(x = drv, fill = factor(cyl))) +
geom_bar(position = "fill") +
labs(
title = "Proportion of cylinder groups within each drive type",
x = "drv",
y = "Proportion within drive type",
fill = "factor(cyl)"
) +
theme_bw()
print(p_b_prop)
ggsave("HW3_output/p_b_prop.png", p_b_prop, width = 7, height = 5, dpi = 300)
The proportion plot makes each bar total 1. This shows the mix of
cylinder groups within each drive type, not the number of vehicles.
The original counts can differ even though the proportion bars have equal heights.
50:1 (Top Level) R Script
R 4.6.1 · ~/
> print(p_b_count)
> ggsave("HW3_output/p_b_count.png", p_b_count, width = 7, height = 5, dpi = 300)
> p_b_prop <- ggplot(vehicles, aes(x = drv, fill = factor(cyl))) +
+ geom_bar(position = "fill") +
+ labs(
+ title = "Proportion of cylinder groups within each drive type",
+ x = "drv",
+ y = "Proportion within drive type",
+ fill = "factor(cyl)"
+) +
+ theme_bw()
> print(p_b_prop)
> ggsave("HW3_output/p_b_prop.png", p_b_prop, width = 7, height = 5, dpi = 300)
Environment History Connections Tutorial
Import Dataset 656 MiB
Global Environment
n_y 40
name_candidates chr [1:5] "community" "area_name" "name" "community_area_name" "pri…
name_field "branch_"
original_rows 6L
out_dir "HW3_outputs"
overall_mean_c 20.3571428571429
p_a Large ggplot2::ggplot (16.7 MB)
p_A Large ggplot2::ggplot (16.7 MB)
p_b_count Large ggplot2::ggplot (16.7 MB)
p_B_count Large ggplot2::ggplot (16.7 MB)
p_b_prop Large ggplot2::ggplot (16.7 MB)
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Proportion of cylinder groups within each drive type
Proportion within drive type
drv
factor(cyl)
4
5
6
8

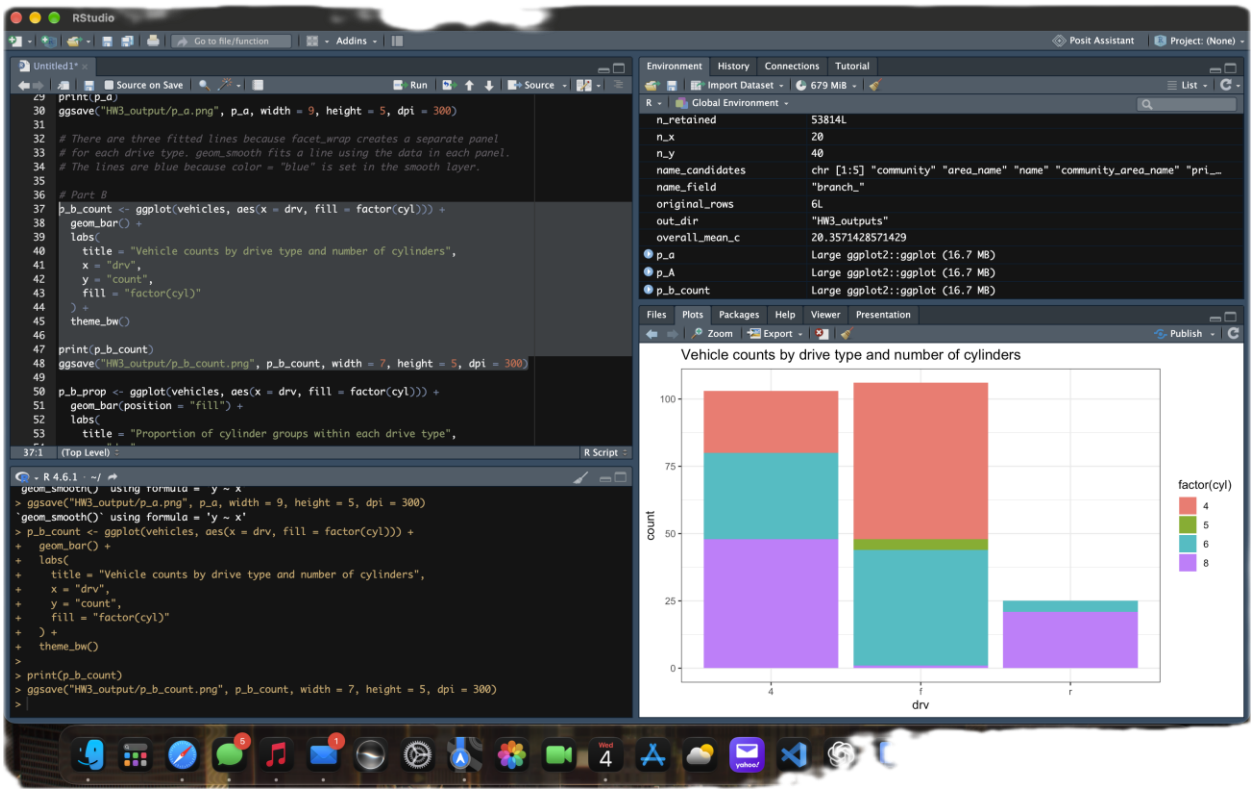

RStudio
Untitled1*
x = "Carat",
y = "Count",
subtitle = "Bin width = 0.01"
) +
theme_bw()
p_coarse <- ggplot(gems_small, aes(x = carat)) +
geom_histogram(binwidth = 0.5, boundary = 0, closed = "right",
color = "white") +
coord_cartesian(xlim = c(0, 2.5)) +
labs(
title = "Diamond carat distribution (coarse bins)",
x = "Carat",
y = "Count",
subtitle = "Bin width = 0.5"
) +
theme_bw()
print(p_fine)
print(p_coarse)
ggsave("HW3_output/p_fine.png", p_fine, width = 8, height = 5, dpi = 300)
ggsave("HW3_output/p_coarse.png", p_coarse, width = 8, height = 5, dpi = 300)
carat_counts <- gems %>%
filter(carat %in% c(0.99, 1.00)) %>%
R 4.6.1 · ~/
Diamond carat distribution (fine bins)
Bin width = 0.01
Count
Carat

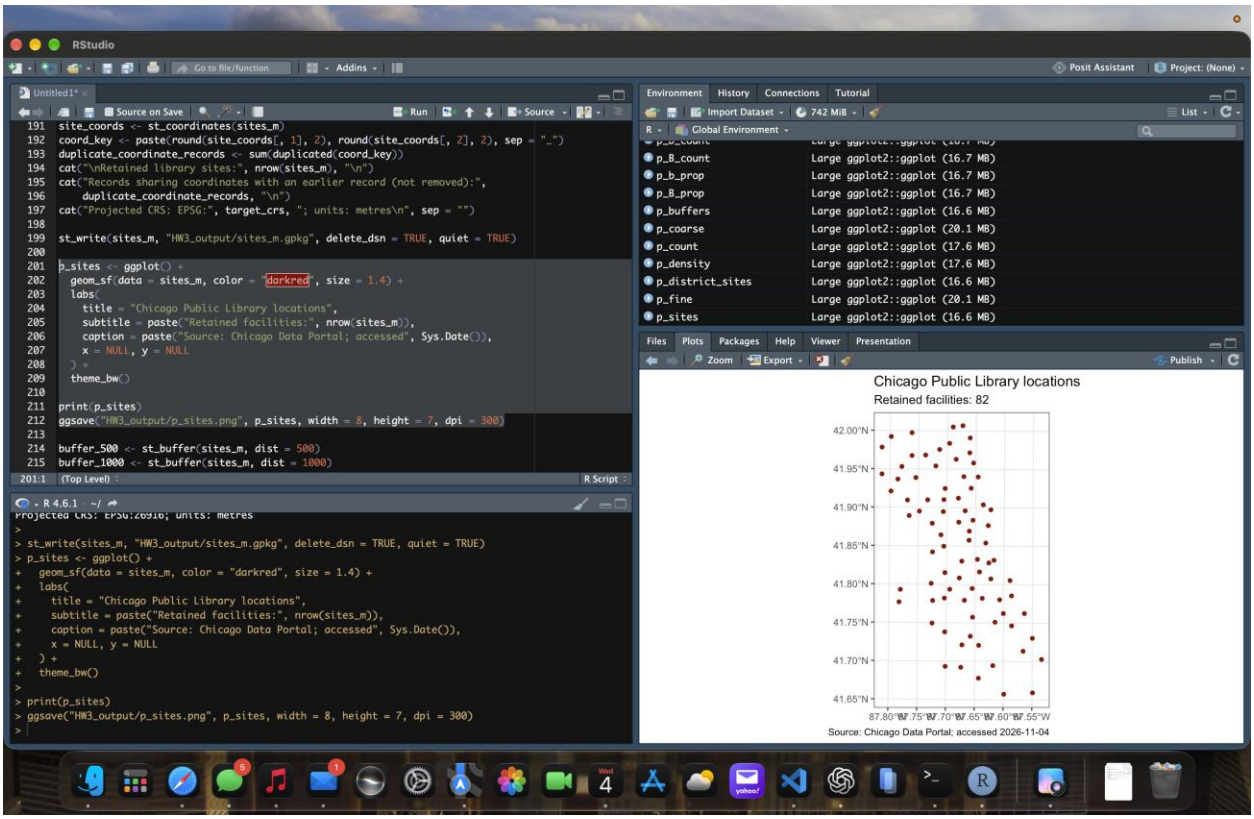

RStudio
Untitled1*
x = "Carat",
y = "Count",
subtitle = "Bin width = 0.01"
) +
theme_bw()
p_coarse <- ggplot(gems_small, aes(x = carat)) +
geom_histogram(binwidth = 0.5, boundary = 0, closed = "right",
color = "white") +
coord_cartesian(xlim = c(0, 2.5)) +
labs(
title = "Diamond carat distribution (coarse bins)",
x = "Carat",
y = "Count",
subtitle = "Bin width = 0.5"
) +
theme_bw()
print(p_fine)
print(p_coarse)
ggsave("HW3_output/p_fine.png", p_fine, width = 8, height = 5, dpi = 300)
ggsave("HW3_output/p_coarse.png", p_coarse, width = 8, height = 5, dpi = 300)
carat_counts <- gems %>%
filter(carat %in% c(0.99, 1.00)) %>%
R 4.6.1 · ~/
Diamond carat distribution (coarse bins)
Bin width = 0.5
Count
Carat

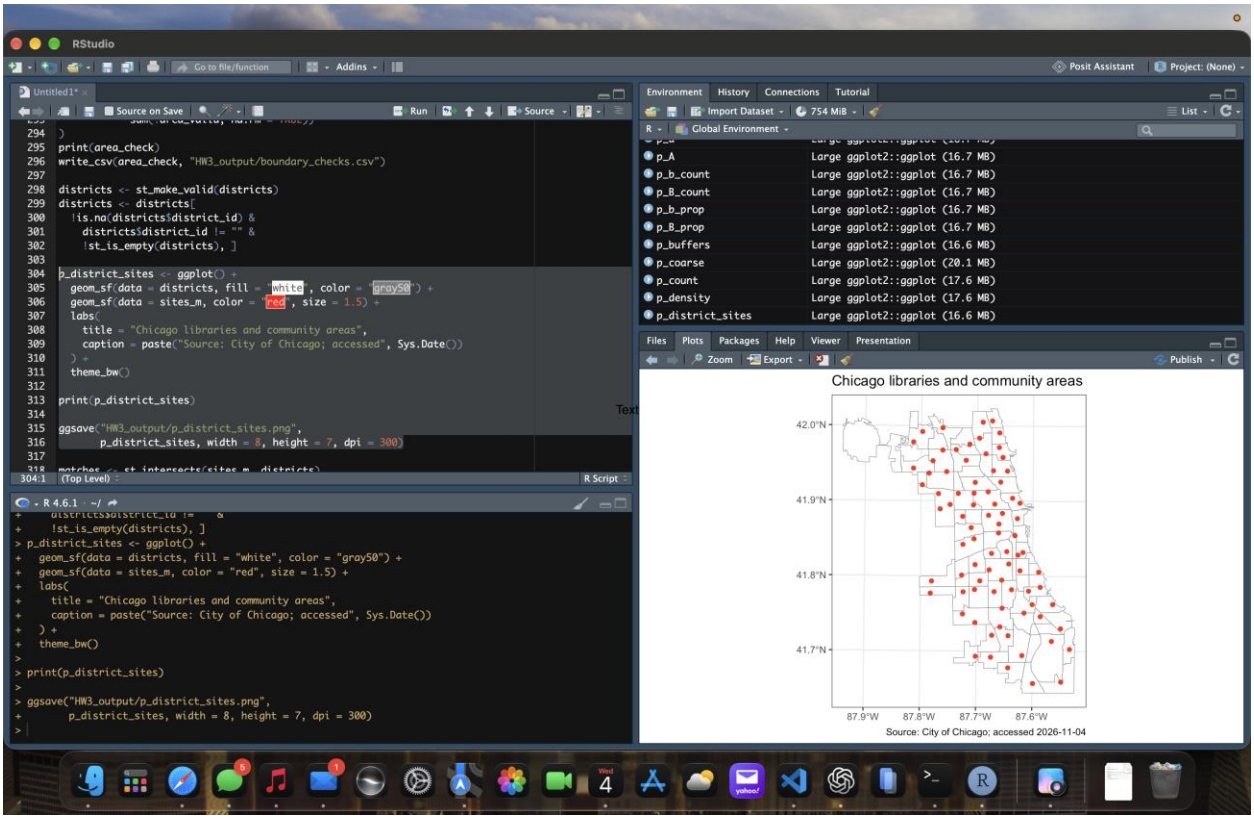

RStudio
Untitled1*
print(p_sites)
ggsave("HW3_output/p_sites.png", p_sites, width = 8, height = 7, dpi = 300)
buffer_500 <- st_buffer(sites_m, dist = 500)
buffer_1000 <- st_buffer(sites_m, dist = 1000)
union_500 <- st_union(buffer_500)
union_1000 <- st_union(buffer_1000)
p_buffers <- ggplot() +
geom_sf(data = st_as_sf(st_sfc(union_500, crs = target_crs)),
fill = "skyblue", color = "steelblue", alpha = 0.35) +
geom_sf(data = sites_m, color = "darkred", size = 1.3) +
labs(
title = "Chicago libraries and combined 500-metre buffers",
caption = paste("Source: Chicago Data Portal; accessed", Sys.Date()),
x = NULL, y = NULL
) +
theme_bw()
print(p_buffers)
ggsave("HW3_output/p_buffers.png", p_buffers, width = 8, height = 7, dpi = 300)
buffer_summary <- tibble(
R 4.6.1 · ~/
> print(p_buffers)
> ggsave("HW3_output/p_buffers.png", p_buffers, width = 8, height = 7, dpi = 300)
Environment
History
Connections
Tutorial
original_rows 6L
out_dir "HW3_outputs"
overall_mean_c 20.3571428571429
p_a Large ggplot2::ggplot (16.7 MB)
p_A Large ggplot2::ggplot (16.7 MB)
p_b_count Large ggplot2::ggplot (16.7 MB)
p_B_count Large ggplot2::ggplot (16.7 MB)
p_b_prop Large ggplot2::ggplot (16.7 MB)
p_B_prop Large ggplot2::ggplot (16.7 MB)
p_buffers Large ggplot2::ggplot (16.6 MB)
Files
Plots
Packages
Help
Viewer
Presentation
Chicago libraries and combined 500-metre buffers
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.80°W
87.75°W
87.70°W
87.65°W
87.60°W
87.55°W
Source: Chicago Data Portal; accessed 2026-11-04


RStudio
Untitled1*
st_write(district_counts, "HW3_output/district_counts.gpkg",
delete_dsn = TRUE, quiet = TRUE)
write_csv(st_drop_geometry(district_counts), "HW3_output/district_counts.csv")
p_count <- ggplot(district_counts) +
geom_sf(aes(fill = n_sites), color = "white", linewidth = 0.2) +
scale_fill_gradient(
low = "lightyellow", high = "darkred",
name = "Facilities\n(count)"
) +
labs(
title = "Chicago library counts by community area",
caption = paste("Source: City of Chicago; accessed", Sys.Date(),
"| Sites not assigned exactly once:", sum(match_count != 1)),
x = NULL, y = NULL
) +
theme_bw()
print(p_count)
ggsave("HW3_output/p_count.png", p_count, width = 8, height = 7, dpi = 300)
p_density <- ggplot(district_counts) +
geom_sf(aes(fill = sites_per_km2), color = "white", linewidth = 0.2) +
scale_fill_gradient(
R 4.6.1 · ~/
> print(p_count)
> ggsave("HW3_output/p_count.png", p_count, width = 8, height = 7, dpi = 300)
Environment
History
Connections
Tutorial
overall_mean_c 20.3571428571429
p_a Large ggplot2::ggplot (16.7 MB)
p_A Large ggplot2::ggplot (16.7 MB)
p_b_count Large ggplot2::ggplot (16.7 MB)
p_B_count Large ggplot2::ggplot (16.7 MB)
p_b_prop Large ggplot2::ggplot (16.7 MB)
p_B_prop Large ggplot2::ggplot (16.7 MB)
p_buffers Large ggplot2::ggplot (16.6 MB)
p_coarse Large ggplot2::ggplot (20.1 MB)
p_count Large ggplot2::ggplot (17.6 MB)
Files
Plots
Packages
Help
Viewer
Presentation
Chicago library counts by community area
42.0°N
41.9°N
41.8°N
41.7°N
87.9°W
87.8°W
87.7°W
87.6°W
Facilities
(count)
3
2
1
0
Source: City of Chicago; accessed 2026-11-04 | Sites not assigned exactly once: 0


RStudio
Untitled1*
Source on Save
Run
Source
"| Sites not assigned exactly once:", sum(match_count != 1)),
x = NULL, y = NULL
) +
theme_bw()
print(p_count)
ggsave("HW3_output/p_count.png", p_count, width = 8, height = 7, dpi = 300)
p_density <- ggplot(district_counts) +
geom_sf(aes(fill = sites_per_km2), color = "white", linewidth = 0.2) +
scale_fill_gradient(
low = "lightyellow", high = "darkred",
name = "Facilities per\nkm²"
) +
labs(
title = "Chicago library density by community area",
caption = paste("Source: City of Chicago; accessed", Sys.Date(),
"| Sites not assigned exactly once:", sum(match_count != 1)),
x = NULL, y = NULL
) +
theme_bw()
print(p_density)
ggsave("HW3_output/p_density.png", p_density, width = 8, height = 7, dpi = 300)
379:1 (Top Level)
R Script
R 4.6.1 · ~/
> print(p_density)
> ggsave("HW3_output/p_density.png", p_density, width = 8, height = 7, dpi = 300)
Environment
History
Connections
Tutorial
Import Dataset
765 MiB
Global Environment
p_a Large ggplot2::ggplot (16.7 MB)
p_A Large ggplot2::ggplot (16.7 MB)
p_b_count Large ggplot2::ggplot (16.7 MB)
p_B_count Large ggplot2::ggplot (16.7 MB)
p_b_prop Large ggplot2::ggplot (16.7 MB)
p_B_prop Large ggplot2::ggplot (16.7 MB)
p_buffers Large ggplot2::ggplot (16.6 MB)
p_coarse Large ggplot2::ggplot (20.1 MB)
p_count Large ggplot2::ggplot (17.6 MB)
p_density Large ggplot2::ggplot (17.6 MB)
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Chicago library density by community area
42.0°N
41.9°N
41.8°N
41.7°N
87.9°W
87.8°W
87.7°W
87.6°W
Facilities per km²
0.4
0.3
0.2
0.1
0.0
Source: City of Chicago; accessed 2026-11-04 | Sites not assigned exactly once: 0


RStudio
Untitled1*
Source on Save
Run
Source
print(p_a)
ggsave("HW3_output/p_a.png", p_a, width = 9, height = 5, dpi = 300)
There are three fitted lines because facet_wrap creates a separate panel
for each drive type. geom_smooth fits a line using the data in each panel.
The lines are blue because color = "blue" is set in the smooth layer.
Part B
p_b_count <- ggplot(vehicles, aes(x = drv, fill = factor(cyl))) +
geom_bar() +
labs(
title = "Vehicle counts by drive type and number of cylinders",
x = "drv",
y = "count",
fill = "factor(cyl)"
) +
theme_bw()
print(p_b_count)
ggsave("HW3_output/p_b_count.png", p_b_count, width = 7, height = 5, dpi = 300)
p_b_prop <- ggplot(vehicles, aes(x = drv, fill = factor(cyl))) +
geom_bar(position = "fill") +
labs(
title = "Proportion of cylinder groups within each drive type",
37:1 (Top Level)
R Script
R 4.6.1 · ~/
> ggsave("HW3_output/p_a.png", p_a, width = 9, height = 5, dpi = 300)
`geom_smooth()` using formula = 'y ~ x'
> p_b_count <- ggplot(vehicles, aes(x = drv, fill = factor(cyl))) +
+ geom_bar() +
+ labs(
+ title = "Vehicle counts by drive type and number of cylinders",
+ x = "drv",
+ y = "count",
+ fill = "factor(cyl)"
+) +
+ theme_bw()
> print(p_b_count)
> ggsave("HW3_output/p_b_count.png", p_b_count, width = 7, height = 5, dpi = 300)
Environment
History
Connections
Tutorial
Import Dataset
679 MiB
Global Environment
n_retained 53814L
n_x 20
n_y 40
name_candidates chr [1:5] "community" "area_name" "name" "community_area_name" "pri…
name_field "branch_"
original_rows 6L
out_dir "HW3_outputs"
overall_mean_c 20.3571428571429
p_a Large ggplot2::ggplot (16.7 MB)
p_A Large ggplot2::ggplot (16.7 MB)
p_b_count Large ggplot2::ggplot (16.7 MB)
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Vehicle counts by drive type and number of cylinders
count
100
75
50
25
0
4
f
r
drv
factor(cyl)
4
5
6
8


RStudio
Untitled1*
Source on Save
Run
Source
site_coords <- st_coordinates(sites_m)
coord_key <- paste(round(site_coords[, 1], 2), round(site_coords[, 2], 2), sep = "_")
duplicate_coordinate_records <- sum(duplicated(coord_key))
cat("\nRetained library sites:", nrow(sites_m), "\n")
cat("Records sharing coordinates with an earlier record (not removed):",
duplicate_coordinate_records, "\n")
cat("Projected CRS: EPSG:", target_crs, "; units: metres\n", sep = "")
st_write(sites_m, "HW3_output/sites_m.gpkg", delete_dsn = TRUE, quiet = TRUE)
p_sites <- ggplot() +
geom_sf(data = sites_m, color = "darkred", size = 1.4) +
labs(
title = "Chicago Public Library locations",
subtitle = paste("Retained facilities:", nrow(sites_m)),
caption = paste("Source: Chicago Data Portal; accessed", Sys.Date()),
x = NULL, y = NULL
) +
theme_bw()
print(p_sites)
ggsave("HW3_output/p_sites.png", p_sites, width = 8, height = 7, dpi = 300)
buffer_500 <- st_buffer(sites_m, dist = 500)
buffer_1000 <- st_buffer(sites_m, dist = 1000)
201:1 (Top Level)
R Script
R 4.6.1 · ~/
Projected CRS: EPSG:26916; units: metres
> st_write(sites_m, "HW3_output/sites_m.gpkg", delete_dsn = TRUE, quiet = TRUE)
> p_sites <- ggplot() +
+ geom_sf(data = sites_m, color = "darkred", size = 1.4) +
+ labs(
+ title = "Chicago Public Library locations",
+ subtitle = paste("Retained facilities:", nrow(sites_m)),
+ caption = paste("Source: Chicago Data Portal; accessed", Sys.Date()),
+ x = NULL, y = NULL
+) +
+ theme_bw()
> print(p_sites)
> ggsave("HW3_output/p_sites.png", p_sites, width = 8, height = 7, dpi = 300)
Environment
History
Connections
Tutorial
Import Dataset
742 MiB
List
R
Global Environment
p_B_count Large ggplot2::ggplot (16.7 MB)
p_b_prop Large ggplot2::ggplot (16.7 MB)
p_B_prop Large ggplot2::ggplot (16.7 MB)
p_buffers Large ggplot2::ggplot (16.6 MB)
p_coarse Large ggplot2::ggplot (20.1 MB)
p_count Large ggplot2::ggplot (17.6 MB)
p_density Large ggplot2::ggplot (17.6 MB)
p_district_sites Large ggplot2::ggplot (16.6 MB)
p_fine Large ggplot2::ggplot (20.1 MB)
p_sites Large ggplot2::ggplot (16.6 MB)
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Chicago Public Library locations
Retained facilities: 82
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.80°W
87.75°W
87.70°W
87.65°W
87.60°W
87.55°W
Source: Chicago Data Portal; accessed 2026-11-04


RStudio
Untitled1*
Source on Save
Run
Source
print(area_check)
write_csv(area_check, "HW3_output/boundary_checks.csv")
districts <- st_make_valid(districts)
districts <- districts[
!is.na(districts$district_id) &
districts$district_id != "" &
!st_is_empty(districts),]
p_district_sites <- ggplot() +
geom_sf(data = districts, fill = "white", color = "gray50") +
geom_sf(data = sites_m, color = "red", size = 1.5) +
labs(
title = "Chicago libraries and community areas",
caption = paste("Source: City of Chicago; accessed", Sys.Date())
) +
theme_bw()
print(p_district_sites)
ggsave("HW3_output/p_district_sites.png",
p_district_sites, width = 8, height = 7, dpi = 300)
304:1 (Top Level)
R Script
R 4.6.1 · ~/
> p_district_sites <- ggplot() +
+ geom_sf(data = districts, fill = "white", color = "gray50") +
+ geom_sf(data = sites_m, color = "red", size = 1.5) +
+ labs(
+ title = "Chicago libraries and community areas",
+ caption = paste("Source: City of Chicago; accessed", Sys.Date())
+) +
+ theme_bw()
> print(p_district_sites)
> ggsave("HW3_output/p_district_sites.png",
+ p_district_sites, width = 8, height = 7, dpi = 300)
Environment
History
Connections
Tutorial
Import Dataset
754 MiB
List
R
Global Environment
p_A Large ggplot2::ggplot (16.7 MB)
p_b_count Large ggplot2::ggplot (16.7 MB)
p_B_count Large ggplot2::ggplot (16.7 MB)
p_b_prop Large ggplot2::ggplot (16.7 MB)
p_B_prop Large ggplot2::ggplot (16.7 MB)
p_buffers Large ggplot2::ggplot (16.6 MB)
p_coarse Large ggplot2::ggplot (20.1 MB)
p_count Large ggplot2::ggplot (17.6 MB)
p_density Large ggplot2::ggplot (17.6 MB)
p_district_sites Large ggplot2::ggplot (16.6 MB)
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Chicago libraries and community areas
42.0°N
41.9°N
41.8°N
41.7°N
87.9°W
87.8°W
87.7°W
87.6°W
Source: City of Chicago; accessed 2026-11-04